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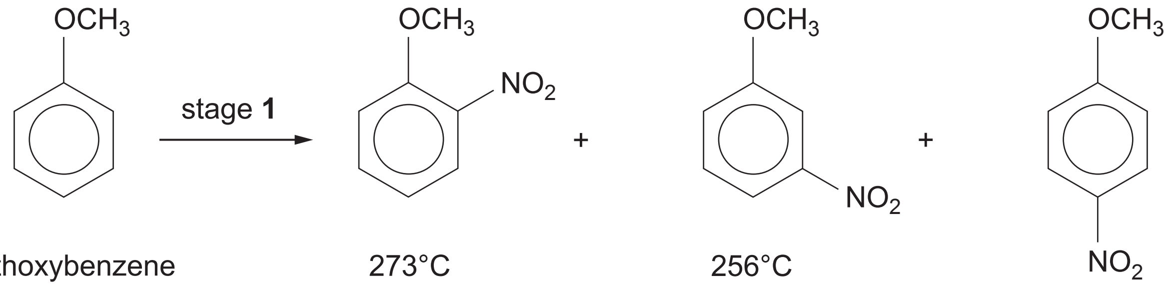
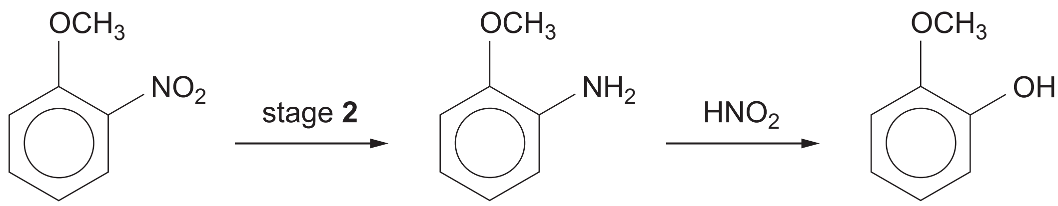
9. (a) Guaiacol (2-methoxyphenol) can be produced in a three-stage method from anisole (methoxybenzene). The boiling temperature of each product of stage **1** is shown.



methoxybenzene 273°C 256°C 260°C



guaiacol

Use your knowledge of a similar reaction, using benzene as the starting material, to answer parts (i)-(iii).

(i) Suggest a reagent(s) for stage **1**. [1]

....

(ii) Suggest a reagent(s) for stage **2**. [1]

....

(iii) Suggest and explain a practical problem that might occur at the end of stage **1**, before proceeding to stage **2**. [1]

....

....

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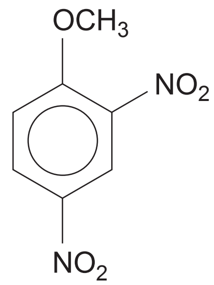
(iv) Complete the table below, indicating what is observed (if anything) when guaiacol reacts with aqueous solutions of these reagents. [2]

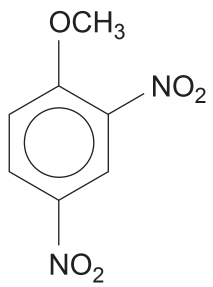
Reagent	$FeCl_3$	$NaHCO_3$
Observation		

(b) 2,4-Dinitroanisole can also be made by the nitration of anisole.



This compound has been used as an explosive. On detonation, it gives carbon, carbon monoxide, nitrogen and steam.

Balance the equation for this reaction. [1]



$\longrightarrow$ .... C + ... CO + ... N_2 + ... H_2O

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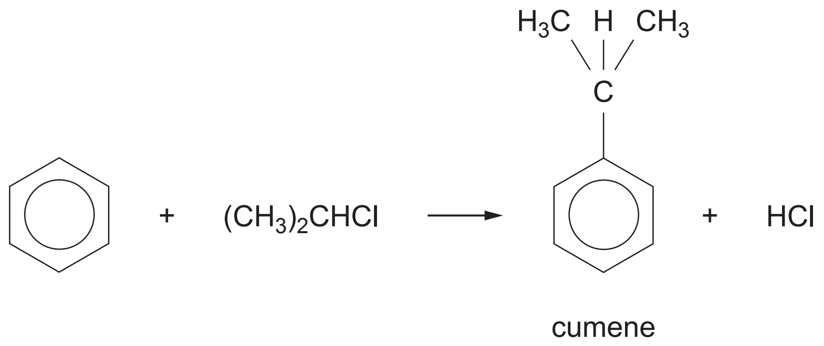
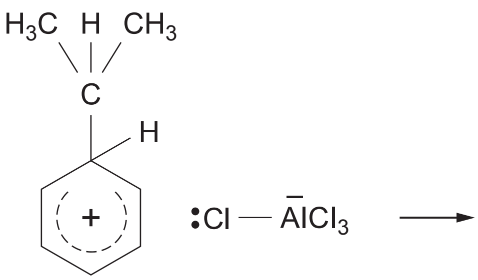
(c) (1-Methylethyl)benzene (commonly called cumene) can be prepared from benzene by a Friedel-Crafts reaction.



The products of the first step of the reaction are given below.

Complete the mechanism, showing how cumene, hydrogen chloride and the catalyst are formed from the intermediate product. [2]



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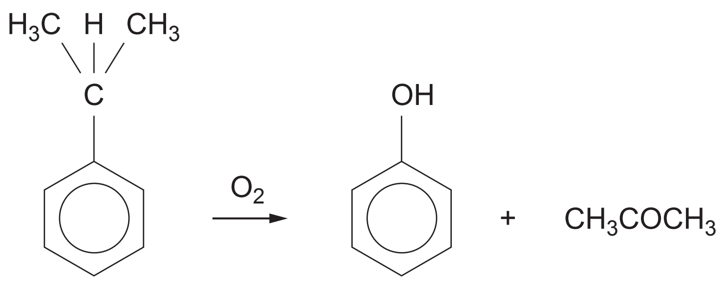
(d) Cumene is an important material in the industrial preparation of phenol and propanone.



(i) The yield of phenol (M_r 94) in this process is 86% with benzene (M_r 78) as the starting material.

Calculate the mass of phenol produced from 234 kg of benzene. [2]

Mass of phenol = ... kg

(ii) During this reaction a cumene radical is produced.

State what is meant by the term 'radical'. [1]

....

....

(iii) Give the formula of a radical formed during the reaction of methane with chlorine. [1]

....

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(e) Phenol reacts with aqueous bromine to give 2,4,6-tribromophenol.

If aqueous bromine is added slowly to an aqueous solution of phenol, describe how you would know when the reaction is just complete. [1]

....

....

(f) Phenol reacts with benzoyl chloride, to give the ester phenyl benzoate and hydrogen chloride.

(i) Write the equation for this reaction. [1]

(ii) The reaction in part (i) is carried out in the presence of aqueous sodium hydroxide, which removes HCl as it is formed.

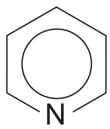
Explain why this method would not be suitable for a similar reaction, using ethanoyl chloride in place of benzoyl chloride. [1]

....

....

(iii) Esterification reactions, such as those in part (i), are sometimes carried out using pyridine in place of sodium hydroxide.

Explain the feature present in the structure of pyridine that enables it to react in this way. [2]



....

....

....

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